GABRIELLA'S MATH LAB · DECIMALS WORKBOOK

SECTION 8

Mixed Decimal Practice

Put it all together: choose the operation, solve, and check. Part A is the Georgia Milestones core review.

SKILL
Choose the operation.

SHOW HOW
Read for clue words.

PART A
Milestones core review.

PART B
× and ÷ review (supplemental).

8.1	Choosing the Operation	☐ done	2
A	Georgia Milestones Core Review (5.NR.4)	☐ done	4
B	Supplemental Review: Multiplying and Dividing	☐ done	8
🔑	Parent Answer Key		10

FOR PARENTS

Part A covers what the Georgia Grade 5 decimal standard asks for: place value, reading and writing, comparing and ordering to thousandths, rounding to hundredths, and adding and subtracting to hundredths, in the question styles used on tests (multiple choice, fill in, and word problems). **Part B** reviews multiplying and dividing decimals, which Georgia teaches in Grade 6. Score the two parts separately.

Name _____ Started _____

Georgia K-12 Mathematics Standards 5.NR.1.1 and 5.NR.4.1–5.NR.4.4 (Part A). Part B is supplemental (6.NR.1).

8.1 Choosing the Operation

Goal: I can choose +, −, ×, or ÷ from the story. Mixes core (+ −) and supplemental (× ÷) problems.

SHOW HOW

Gabriella has $10.00. She buys a book for $6.45. How much money does she have left?

1. **Find the question:** "How much … left?" — something is taken away.
2. **Choose the operation** with the clue words below: "left" means **subtract**.
3. **Estimate:** 10 − 6 = about $4. **Solve:** 10.00 − 6.45 = **3.55**.
4. **Check and label:** 3.55 + 6.45 = 10.00 ✓. She has **$3.55** left.

Add	in all, total, altogether, combined, sum, how much … together
Subtract	left, remain, how much more / less, how much taller / longer, difference, change
Multiply	each (and you want the total), per, equal groups of, times as much, area
Divide	share equally, split, each (and you know the total), how many … fit, per

Why it works: the operation comes from the **story**, not the numbers. Ask: am I joining, taking away, making equal groups, or splitting into equal groups?

LET'S TRY ONE

One bag of popcorn costs $1.25. How much do 4 bags cost?

Clue: _____ Operation: _____

Answer with units: _____

YOUR TURN — SKILL 8.1 · WRITE THE OPERATION, THEN SOLVE

1 Leo walked **2.75 km** in the morning and **1.5 km** after lunch. How far did he walk in all?

Operation and answer _____ km

2 A jug has **3.2 L** of milk. The family drinks **1.85 L**. How much milk is left?

Operation and answer _____ L

3 One eraser costs **$0.45**. What do **6** erasers cost?

Operation and answer _____ $

4 Three sisters share **8.4 m** of ribbon equally. How much does each get?

Operation and answer _____ m

5 Mia is **1.52 m** tall. Her brother is **1.38 m** tall. How much taller is Mia?

Operation and answer _____ m

6 A puzzle costs **$12.99** and a book costs **$4.50**. What is the total cost?

Operation and answer _____ $

7 A bottle holds **6 L**. A glass holds **0.75 L**. How many glasses can be filled?

Operation and answer _____ glasses

8 A garden bed is **2.5 m** long and **3.2 m** wide. What is its area in square meters?

Operation and answer _____ square meters

PART A · Georgia Milestones Core Review

5.NR.1.1 · 5.NR.4 · place value, compare, order, round, add, subtract

YOUR TURN — **PART A · PROBLEMS 1–8**

1 What is the value of the **8** in **6.384**?

(A) **8** (B) **0.8** (C) **0.08** (D) **0.008**

2 Write **5.207** in expanded form.

Expanded _____

3 Write **"three and forty-five thousandths"** as a decimal.

Decimal _____

4 The 6 in **0.6** is how many times the value of the 6 in **0.006**?

Answer _____ times

5 Write >, <, or =: **0.47** ☐ **0.5**

6 Write >, <, or =:

2.080 ☐ **2.08**

7 Order from least to greatest:

0.62, 0.602, 0.6, 0.626

Least → greatest _____

8 Round **7.846** to the nearest hundredth.

(A) **7.8** (B) **7.84** (C) **7.85** (D) **7.9**

YOUR TURN — PART A · PROBLEMS 9–16

9 Round **12.5** to the nearest whole number.

Rounds to _____

10 Round **0.952** to the nearest tenth.

Rounds to _____

11 **4.56 + 3.78**

Sum _____

12 **7.4 + 2.65**

Sum _____

13 **9.2 − 4.75**

Difference _____

14 **6 − 3.28**

Difference _____

15 Which is the sum of **3.05 + 0.6**?

(A) **3.11** (B) **3.65** (C) **3.56**

(D) **9.05**

16 Which is the difference **5.4 − 2.38**?

(A) **3.02** (B) **3.14** (C) **2.02**

(D) **3.12**

YOUR TURN — PART A · WORD PROBLEMS 17–22

17 In a race, Ava's time was **4.25 seconds** and Jo's was **3.8 seconds**. How many seconds faster was Jo? Add or subtract?

Operation and answer _____ seconds

18 Dad jogged **1.75 miles** before lunch and **2.4 miles** after. How far did he jog in all?

Answer _____ miles

19 Gabriella pays for a **$14.37** game with a **$20** bill. How much change does she get?

$ _____

20 A melon weighs **3.468 kg**. What is its weight to the nearest tenth of a kilogram?

About _____ kg

21 Three ribbons are **1.09 m**, **1.9 m**, and **1.19 m** long. Order them from longest to shortest.

Longest → shortest _____

22 Mia has **$8.00**. She wants a **$3.45** sandwich, a **$2.79** juice, and a **$1.65** apple. Does she have enough money? Show how you know.

Answer _____

YOUR TURN — PART A · EXPLAIN YOUR THINKING, 23–24

23 Sam says **0.75 is greater than 0.8** because 75 is greater than 8. Is Sam correct? Explain using place value or a model.

Which is greater? _____

24 A tank has **4.2 L** of water. **1.55 L** leaks out. How much water is left? Show your work, then check your answer with addition.

Answer _____

PART A SCORE

Correct: _____ out of 24. Skills to practice again (circle): **place value · reading/writing · comparing · ordering · rounding · adding · subtracting · word problems**

PART B · Supplemental Review: Multiplying and Dividing

Grade 6 (6.NR.1); #1–2 review 5.NR.1.2

YOUR TURN **PART B · PROBLEMS 1–8**

1 0.56 × 100

Product _____

2 3.7 ÷ 10

Quotient _____

3 3.4 × 5

Product _____

4 0.6 × 0.7

Product _____

5 1.5 × 2.4

Product _____

6 8.75 ÷ 5

Quotient _____

7 9 ÷ 4

Quotient _____

8 2.8 ÷ 0.4

Quotient _____

YOUR TURN — PART B · WORD PROBLEMS 9–14

9 Notebooks cost **$2.35** each. How much do **3** notebooks cost?

$ _____

10 A **7.2 m** rope is cut into **6** equal pieces. How long is each piece?

Answer _____ m

11 A pitcher holds **1.6 L** when full. It is half (**0.5**) full. How many liters are in it?

Answer _____ L

12 Dad has **4.5 kg** of soil. Each bag holds **0.9 kg**. How many bags can he fill?

Answer _____ bags

13 Estimate **5.9 × 4.1** by rounding to whole numbers.

Estimate _____

14 Five friends share a **$9.00** pizza equally. Which operation? How much does each pay? Check with the inverse operation.

Answer _____

PART B SCORE

Correct: _____ out of 14. These are mostly Grade 6 skills: a score below half just means more time with Sections 6 and 7.

Parent Answer Key

Equal values are correct (3.50 = 3.5). For 8.1, both the operation and the answer must be right. "Try" is the Let's Try One problem.

8.1 Choosing the Operation

Try **Multiply: $5.00** 4 equal groups of $1.25 (we want the total): multiply. 125 × 4 = 500, 2 decimal places → 5.00: $5.00.

1 **Add: 4.25 km** Line up the points; write 2.75 and 1.50. Hundredths: 5 + 0 = 5. Tenths: 7 + 5 = 12, write 2, carry 1. Ones: 2 + 1 + 1 (carried) = 4. → 4.25.

2 **Subtract: 1.35 L** Line up the points; write 3.20 − 1.85. Hundredths: 0 < 5, regroup: 10 − 5 = 5. Tenths (2 became 1): 1 < 8, regroup: 11 − 8 = 3. Ones (3 became 2): 2 − 1 = 1. → 1.35.

3 **Multiply: $2.70** Multiply as whole numbers: 45 × 6 = 270. Decimal places: 2 + 0 = 2, so 270 → 2.70.

4 **Divide: 2.8 m** Long division: 8 ÷ 3 = 2 r 2; point goes straight up; 24 ÷ 3 = 8. So 8.4 ÷ 3 = 2.8. Check: 2.8 × 3 = 8.4.

5 **Subtract: 0.14 m** Line up the points. Hundredths: 2 < 8, regroup: 12 − 8 = 4. Tenths (5 became 4): 4 − 3 = 1. Ones: 1 − 1 = 0. → 0.14.

6 **Add: $17.49** Line up the points; write 12.99 and 4.50. Hundredths: 9 + 0 = 9. Tenths: 9 + 5 = 14, write 4, carry 1. Ones: 2 + 4 + 1 (carried) = 7. Tens: 1 + 0 = 1. → 17.49.

7 **Divide: 8 glasses** Multiply both numbers by 100: 6 ÷ 0.75 = 600 ÷ 75 = 8. Check: 8 × 0.75 = 6.

8 **Multiply: 8 square meters** Multiply as whole numbers: 25 × 32 = 800. Decimal places: 1 + 1 = 2, so 800 → 8.00 = 8.

A Part A: Georgia Milestones Core Review

1 **C (0.08)** The 8 is in the hundredths place: 8 hundredths = 0.08.

2 **5 + 0.2 + 0.007** 5 ones, 2 tenths, 7 thousandths (the 0 in the hundredths place adds nothing).

3 **3.045** Before "and": 3. Thousandths = 3 places after the point; 45 has 2 digits, so write a 0 first → 3.045.

4 **100** 6 tenths vs 6 thousandths: the digit is 2 places to the left, so 10 × 10 = 100 times as much.

5 **<** Write 0.47 and 0.50. First different place: tenths, 4 < 5. So 0.47 < 0.5.

6 **=** 2.080 = 2.08: written with the same number of places, the digits are all the same.

7 **0.6, 0.602, 0.62, 0.626** With 3 decimal places: 0.600 < 0.602 < 0.620 < 0.626.

8 **C (7.85)** Rounding digit: 4 (hundredths). Digit to its right: 6, 5 or more, so round up → 7.85.

9 **13** Rounding digit: 2 (ones). Digit to its right: 5, 5 or more, so round up → 13.

10 **1.0** Rounding digit: 9 (tenths). Digit to its right: 5, 5 or more, so round up → 1.0.

11 **8.34** Line up the points. Hundredths: 6 + 8 = 14, write 4, carry 1. Tenths: 5 + 7 + 1 (carried) = 13, write 3, carry 1. Ones: 4 + 3 + 1 (carried) = 8. → 8.34.

12 **10.05** Line up the points; write 7.40 and 2.65. Hundredths: 0 + 5 = 5. Tenths: 4 + 6 = 10, write 0, carry 1. Ones: 7 + 2 + 1 (carried) = 10, write 0, carry 1. Write the carried 1 in the next place on the left. → 10.05.

13 **4.45** Line up the points; write 9.20 − 4.75. Hundredths: 0 < 5, regroup: 10 − 5 = 5. Tenths (2 became 1): 1 < 7, regroup: 11 − 7 = 4. Ones (9 became 8): 8 − 4 = 4. → 4.45.

14 **2.72** Line up the points; write 6.00 − 3.28. Hundredths: 0 < 8, regroup: 10 − 8 = 2. Tenths: the 0 was regrouped to 10, then gave 1 away, so it is 9: 9 − 2 = 7. Ones (6 became 5): 5 − 3 = 2. → 2.72.

15 **B (3.65)** Line up the points; write 3.05 and 0.60. Hundredths: 5 + 0 = 5. Tenths: 0 + 6 = 6. Ones: 3 + 0 = 3. → 3.65.

16 **A (3.02)** Line up the points; write 5.40 − 2.38. Hundredths: 0 < 8, regroup: 10 − 8 = 2. Tenths (4 became 3): 3 − 3 = 0. Ones: 5 − 2 = 3. → 3.02.

A2 Part A (continued): Word Problems and Explaining

17 **Subtract: 0.45 seconds** Line up the points; write 4.25 − 3.80. Hundredths: 5 − 0 = 5. Tenths: 2 < 8, regroup: 12 − 8 = 4. Ones (4 became 3): 3 − 3 = 0. → 0.45.

18 **4.15 miles** Line up the points; write 1.75 and 2.40. Hundredths: 5 + 0 = 5. Tenths: 7 + 4 = 11, write 1, carry 1. Ones: 1 + 2 + 1 (carried) = 4. → 4.15.

19 **$5.63** Line up the points; write 20.00 − 14.37. Hundredths: 0 < 7, regroup: 10 − 7 = 3. Tenths: the 0 was regrouped to 10, then gave 1 away, so it is 9: 9 − 3 = 6. Ones: the 0 was regrouped to 10, then gave 1 away, so it is 9: 9 − 4 = 5. Tens (2 became 1): 1 − 1 = 0. → 5.63.

20 **3.5 kg** Rounding digit: 4 (tenths). Digit to its right: 6, 5 or more, so round up → 3.5.

21 **1.9, 1.19, 1.09** With 2 decimal places: 1.90 > 1.19 > 1.09.

22 **Yes, she has enough ($0.11 left)** Total: 3.45 + 2.79 + 1.65 = 7.89. 8.00 − 7.89 = 0.11. She has enough, with $0.11 left.

23 **0.8 is greater** Write 0.8 as 0.80. Compare tenths: 8 > 7, so 0.80 > 0.75 (80 hundredths > 75 hundredths). Sam compared 8 and 75 as whole numbers, but the digits are in different places.

24 **2.65 L** Line up the points: 4.20 − 1.55. Hundredths: 0 < 5, regroup: 10 − 5 = 5. Tenths: 1 < 5, regroup: 11 − 5 = 6. Ones: 3 − 1 = 2. → 2.65 L. Check: 2.65 + 1.55 = 4.20. Common mistake: lining up 4.2 under 1.55 by the right edge.

B Part B: Supplemental Review (× and ÷)

1 **56** Multiplying by 100: each digit moves 2 places to the left (the point moves 2 right): 0.56 → 56.

2 **0.37** Dividing by 10: each digit moves 1 place to the right (the point moves 1 left): 3.7 → 0.37.

3 **17** Multiply as whole numbers: 34 × 5 = 170. Decimal places: 1 + 0 = 1, so 170 → 17.0 = 17.

4 **0.42** Multiply as whole numbers: 6 × 7 = 42. Decimal places: 1 + 1 = 2, so 42 → 0.42.

5 **3.6** Multiply as whole numbers: 15 × 24 = 360. Decimal places: 1 + 1 = 2, so 360 → 3.60 = 3.6.

6 **1.75** Long division: 8 ÷ 5 = 1 r 3; point goes straight up; 37 ÷ 5 = 7 r 2; 25 ÷ 5 = 5. So 8.75 ÷ 5 = 1.75. Check: 1.75 × 5 = 8.75.

7 **2.25** Write 9 as 9.00 so you can keep dividing. Long division: 9 ÷ 4 = 2 r 1; point goes straight up; 10 ÷ 4 = 2 r 2; 20 ÷ 4 = 5. So 9 ÷ 4 = 2.25. Check: 2.25 × 4 = 9.

8 **7** Multiply both numbers by 10: 2.8 ÷ 0.4 = 28 ÷ 4 = 7. Check: 7 × 0.4 = 2.8.

9 **$7.05** Multiply as whole numbers: 235 × 3 = 705. Decimal places: 2 + 0 = 2, so 705 → 7.05.

10 **1.2 m** Long division: 7 ÷ 6 = 1 r 1; point goes straight up; 12 ÷ 6 = 2. So 7.2 ÷ 6 = 1.2. Check: 1.2 × 6 = 7.2.

11 **0.8 L** Multiply as whole numbers: 16 × 5 = 80. Decimal places: 1 + 1 = 2, so 80 → 0.80 = 0.8.

12 **5 bags** Multiply both numbers by 10: 4.5 ÷ 0.9 = 45 ÷ 9 = 5. Check: 5 × 0.9 = 4.5.

13 **24** Round each number to the nearest whole: 5.9 ≈ 6, 4.1 ≈ 4. 6 × 4 = 24.

14 **Divide: $1.80 each; check 1.80 × 5 = 9.00** Share equally: divide. 9.00 ÷ 5: long division 9 ÷ 5 = 1 r 4; point goes straight up; 40 ÷ 5 = 8; 0 ÷ 5 = 0 → 1.80. Check: 1.80 × 5 = 9.00.